Fiscal Year:	FY 2018	**Task Last Updated:**	FY 01/22/2018
PI Name:	Cooper, Maya M.S.		
Project Title:	Improvement of Shelf Life for Space Food Through Hurdle Approach		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	03/08/2017	**End Date:**	09/30/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2019 (previously 9/30/2024) due to PI change from Maya Cooper to Takiyah Sirmons in October 2019; project continues as "Improvement of Shelf Life for Space Food Through Hurdle Approach (PI=Sirmons)" (Ed., 12/13/19)		
Key Personnel Changes/Previous PI:	January 2018 report: No personnel changes during reporting period.		
COI Name (Institution):	Sirmons, Takiyah Ph.D. (Leidos/NASA Johnson Space Center) Froio-Blumsack, Danielle M.S. (U.S. Army Natick Soldier RD&E Center) Mohr, Lea B.S. (Ameriqual) Douglas, Grace Ph.D. (NASA Johnson Space Center) Young, Millennia Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Most items of the current space food system will not achieve the minimum 5-year shelf life required for a Mars mission due to decrements in nutritional quality or sensory acceptability. Previous Advanced Food Technology (AFT) studies have shown critical losses in some nutrients in a number of space food products after 3 years of ambient storage (Cooper project, “Effects of Processing and Subsequent Storage on Nutrition (PI Cooper)”), unacceptable losses in quality after 3 years (Catauro, P.M. & Perchonok, M.H. Assessment of the long-term stability of retort pouch foods to support extended duration spaceflight. Journal of food science (2012) 77, S29-39), and the inability of individual processing and storage solutions to achieve a projected 5-year shelf life (Cooper project “Integration of Product, Package, Process, and Environment: A Food System Optimization”). This task will investigate the use of hurdle approach to increase the shelf life of the space food system. The study will produce the 5-year data essential to fully inform the state of a Mars food system and indicate the best countermeasures to nutritional and sensory degradation. Putting a 7-year data option in place initially will ensure that the PRR is not unnecessarily extended by several years if it is determined at that time that a longer-term shelf life study was necessary to determine mission requirements. Specific Aims: Determine how reduced storage temperatures (-80C, -20C, 4C) and alternative processing and packaging impact the quality and nutrient concentrations of space food over a 5-7 year shelf life period as compared to the quality and nutrient concentrations of space food produced under traditional methods and stored at ambient temperature (21C).
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal. The results of this study will influence vehicle design. We have established a collaboration with the Advanced Exploration Systems Logistics Reduction (AES LR) team, who will contribute their own funding to evaluate storage solutions and feasibility of designs. AES will ultimately be responsible for the food storage solution on a mission to Mars, and their location at Johnson Space Center and their collaboration at this initial point will ensure that solutions are integrated successfully in the final vehicle design. Only four storage conditions can be assessed within this experiment, due to funding limitations. The AES team participated in discussions in determining the experimental storage conditions to ensure direct applicability to potential scenarios of food storage for long duration missions. The current AES work in vehicle design, and their collaboration with Advanced Food Technology to finalize storage solutions, is the first reason that we propose that this work be accomplished through directed study.
Research Impact/Earth Benefits:	The discovery of pathways to extend the shelf life of space food is directly transferable to the extension of shelf life of military food, emergency food rations, and commercial food items. Longer food shelf life leads to lower food waste, higher food quality in aging food samples, and an opportunity to increase variety within stored food inventory. The exploration of an alternative technology, specifically microwave-assisted thermal sterilization, provides additional data for the FDA approval of this new processing technology, which drives commercial innovation in the food industry.
Task Progress:	The initial year of the Hurdle Approach study involved several key milestones that enable future study progress as well as the initial production and shelf life initiation of several foods. Sixteen representative foods from the International Space Station (ISS) food system were chosen for production and analysis and will be evaluated initially and at 1, 3, and 5 years, with potential for analysis at 7 years if necessary. The study requires the production and storage of approximately 13,000 packages of space food. Ensuring adequate storage volume at each prescribed storage temperature was the first major milestone of the study. An indoor refrigeration unit, 20-foot walk-in refrigerator/freezer unit, and ultra-cold freezer were procured to have adequate storage volume for the study. The second completed milestone for the project was the identification and procurement of the package for the microwave assisted thermal sterilization (MATS), processing at AmeriQual Foods. The Toppan GL-RD barrier pouch was selected for the MATS processing because of its superior oxygen and water vapor transmission barrier properties as compared to other non-metallic food pouches. The predominant activity, however, for the study in the last year has been the preparation for and the initial production of the study foods. AmeriQual Foods contracted with Washington State University to complete D- and z-value testing of the foods undergoing MATS, where D is time required in the food, at a given temperature, for a ten-fold reduction in the number of organisms, and z is the number of degrees the temperature has to be increased to achieve a tenfold reduction in the D-value. At the Space Food Space Food Research Facility, Rice and Chicken, Italian Vegetables, and Turkey Tetrazzini processing and freeze drying were completed. Strawberry processing and freeze drying was completed at the Space Food Systems Laboratory.
Bibliography Type:	Description: (Last Updated: 04/23/2019)
Abstracts for Journals and Proceedings	Cooper MR, Sirmons TA, Froio-Blumsack D, Mohr L, Young M, Douglas GL. "Extension of Space Food Shelf Life Through Hurdle Approach." Poster presentation at 2018 NASA HRP Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators’ Workshop, Galveston, TX, January 22-25, 2018. Available online at https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20170009936.pdf , Jan-2018